

OMNICART MARKETPLACE PROGRAM

Business Requirements Document

Business scope, customer journeys, policies, operating model, payment objectives, and release success measures for a large-scale ecommerce marketplace.

Project:	OmniCart Commerce Platform with Multi-Gateway Payments
Prepared For:	Product, Engineering, QA, Security, Operations, Finance
Generated:	2026-05-13
Classification:	Synthetic test data for ingestion pipeline validation
Version:	1.0

1. Executive Summary

OmniCart is a hypothetical enterprise ecommerce marketplace with customer experiences comparable to a large online retail platform. The program enables browsing millions of SKUs, personalized discovery, cart and checkout, multi-seller fulfillment, digital wallet support, refunds, promotions, loyalty credits, seller settlement, and payment gateway failover across card, UPI, wallet, BNPL, and net-banking providers.

The payment gateway integration is treated as a tier-0 capability because it directly impacts revenue, customer trust, fraud exposure, seller reconciliation, and regulatory compliance. The business requires resilient authorization, capture, refund, reversal, chargeback, settlement, and ledger reporting flows.

Program Goal: Increase checkout conversion, reduce payment failures, support multiple tenders, enable dynamic gateway routing, and deliver auditable financial traceability from order placement to settlement.

2. Business Objectives

ID	Objective	Target Metric	Business Owner
BO-01	Provide Amazon-like end-to-end shopping experience for web and mobile users.	95th percentile product discovery response under 1.8s	Commerce Product
BO-02	Support payment authorization with card, UPI, wallet, gift card, BNPL, net banking, and COD.	98.5% payment initiation success	Payments Product
BO-03	Reduce failed payments via smart retry and gateway routing.	15% reduction in payment abandonment	Payments Product
BO-04	Provide finance-grade reconciliation and seller settlement reports.	99.95% ledger-to-gateway match rate	Finance Ops
BO-05	Meet PCI-DSS, privacy, fraud, and audit obligations.	Zero raw PAN storage; tokenized payment references only	Security and Compliance

3. In-Scope Capabilities

- Customer account, address book, wish list, recommendations, recently viewed items, product ratings, Q&A, and personalized deals.
- Catalog search, faceted filtering, sponsored products, inventory-aware product detail pages, delivery promise, cart, promotions, and checkout.
- Payment orchestration with gateway selection, authorization, capture, void, refund, partial refund, retry, webhook processing, and reconciliation.
- Fraud screening, velocity checks, risk scoring, device fingerprinting, 3DS challenge support, and order hold/release workflow.
- Seller order split, fulfillment routing, shipment tracking, returns, replacements, chargeback evidence, and settlement adjustment.

4. Key Business Journeys

Journey	Primary Actor	Business Outcome	Critical Payment Touchpoints
Browse to Buy	Guest / Registered Customer	Find product, add to cart, pay, receive order confirmation	Offer validation, tax/shipping calculation, authorization, capture policy
Saved Payment Checkout	Registered Customer	Pay with tokenized saved card or wallet	Token vault lookup, CVV/3DS validation, consent audit
Split Cart Multi-Seller Checkout	Customer	One payment for multiple seller shipments	Single authorization, split ledger allocation, seller settlement
Return and Refund	Customer / Support Agent	Refund customer after return eligibility validation	Refund API, original payment lookup, ledger reversal
Payment Failure Recovery	Customer	Recover from failed authorization without losing cart state	Retry, alternate gateway, alternate tender, failure messaging

5. Business Rules

Rule ID	Rule	Priority	Notes
BR-101	Orders must not be confirmed until payment is authorized or COD eligibility is approved.	Critical	Applies to prepaid and BNPL orders.
BR-102	Payment capture is immediate for digital goods and on-shipment for marketplace physical goods unless seller policy differs.	High	Capture policy configurable by fulfillment type.
BR-103	Gift card and wallet balances must be consumed before external payment if user selects mixed tender.	High	Ledger creates separate tender lines.
BR-104	Duplicate payment callbacks must be idempotent and must not create duplicate captures, refunds, or ledger entries.	Critical	Uses idempotency key and gateway transaction id.
BR-105	Failed payment attempts must preserve cart and order intent for at least 30 minutes.	Medium	Improves recovery and conversion.
BR-106	Refunds must follow the original tender unless compliance rules require manual review.	High	Supports split tender refunds.

6. Stakeholders

Role	Name	Interest	Acceptance Focus
Executive Sponsor	VP Commerce	Revenue growth and trust	Conversion, uptime, customer satisfaction
Payments Product Owner	Priya Raman	Payment success and new tenders	Gateway routing, retries, refunds
Engineering Lead	Arjun Mehta	Architecture and service reliability	API contracts, resiliency, observability
QA Lead	Meera Shah	Risk-based coverage	Functional, negative, security, performance tests
Finance Ops	Ravi Menon	Settlement and reconciliation	Ledger accuracy, reports, exception queues

7. High-Level Business Flow



8. Non-Functional Business Expectations

- Payment initiation p95 below 900ms excluding gateway network latency.
- Checkout availability 99.95% monthly; payment webhook ingestion 99.99% monthly.
- All payment status transitions auditable for seven years.
- Customer-facing failure messages must be actionable without exposing gateway internals.
- System must support campaign sale traffic at 8x normal peak for two hours.